

New Insights into User-Defined Smart Ring Gestures with Implications for Gesture Elicitation Studies

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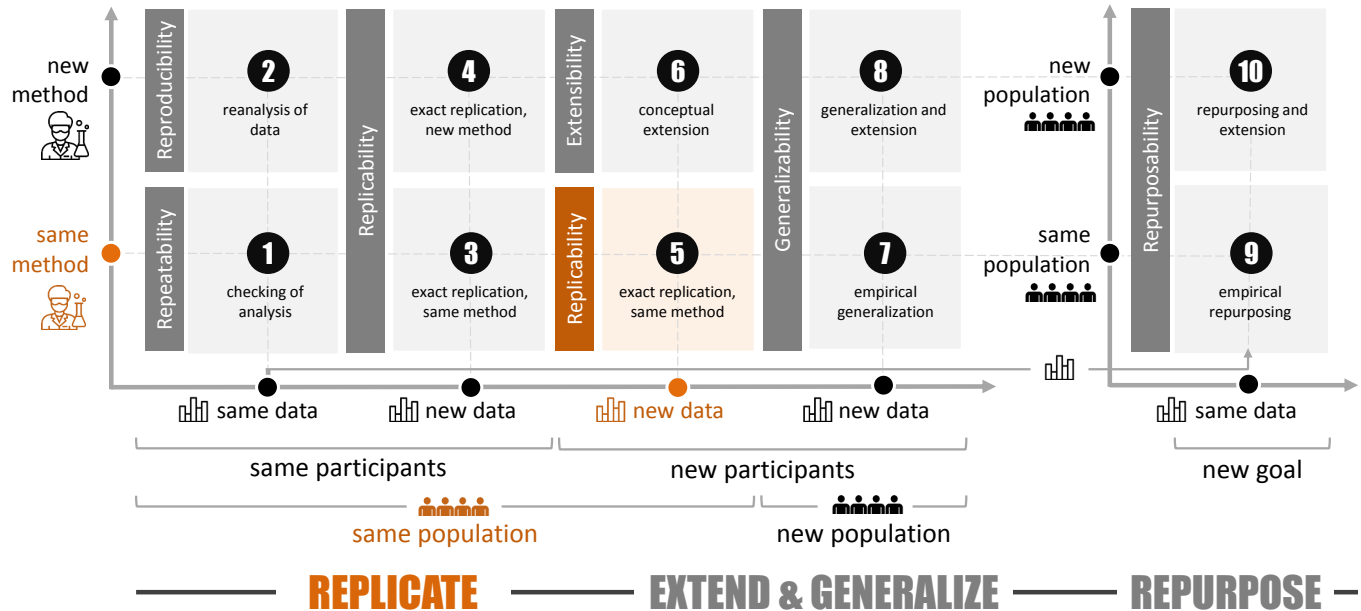


Figure 1: Positioning of the gesture elicitation study conducted in this work in the RepliGES space [14]. Our study is a “type-5 replication,” i.e., it has the same goal and applies the same analysis method as the original, replicated study [11], but with new gestures elicited from new participants of the same population.

ABSTRACT

We conduct a replication of a previously published gesture elicitation study, which collected and examined user-defined gestures performed with smart rings. Our findings reveal that only 50% of the gestures elicited during our replication match those proposed by the participants of the original study, although our sample of participants had the same characteristics as the sample from the original study, and our participants spent the same amount of time to propose gestures and rated the goodness of their gesture-to-function mappings at the same level as the participants of the original study. Furthermore, the new gestures elicited in our replication present

different articulation characteristics in terms of their structure, complexity, symmetry, and locale, compared to the gestures from the original study. Our results have implications for the theory and practice of gesture elicitation studies, for which we provide several recommendations for researchers and practitioners.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI; Gestural input; User studies.

KEYWORDS

Gesture elicitation, replicability, study, smart rings, electronic rings, gesture input, RepliCHI, RepliGES.

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1 INTRODUCTION

Smart rings, as a distinctive type of a finger-augmentation device [26], enable several gesture input modalities [30], such as touch (on the ring surface), motion (by rolling or sliding the ring on the finger), and mid-air input (by drawing symbols in mid-air). However, the vocabulary of gestures used for smart rings is quite limited. For example, Magic Ring [17] employs four gestures only: pointing the finger up, down, left, and right; RingIoT [9] implements pointing, directional movements, and simple geometric shapes, such as a “circle” drawn in mid-air; and Zhang *et al.*’s [40] ring uses six gestures: four directional swipes, “click,” and “circle.” Despite the growing interest in smart rings for interactive systems [7,12,24,27,30,35,41], users’ preferences for ring gestures have been little examined.

In this context, Gheran *et al.* [11] conducted the first study to explore users’ preferences for gestures performed with smart rings. From a dataset of 672 gestures elicited from 24 participants, Gheran *et al.* reported an overall low consensus, but interesting insights about participants’ mental models and articulation characteristics of ring gestures, e.g., 69% of the proposed gestures were performed in mid-air and 42% of the bimanual gestures were asymmetric. To arrive at these results, Gheran *et al.* implemented a gesture elicitation study [37,38], a specialized instance of participatory design [5,20], to unveil users’ preferences for gesture input. In these studies, participants propose gestures they would like to use to effect specific system functions, such as turning off the TV with a smart ring. The practitioner analyzes the elicited gestures to identify the most preferred [32], and to understand which gesture commands are perceived intuitive [37], low effort [38], and memorable [21], likely to reflect the preferences of the larger user population. With time, this understanding should turn into consolidated knowledge [18,33,34] via repeated validation and replication, on which the community can capitalize to inform gesture input design. However, a recent paper [14] revealed very few replications of gesture elicitation studies despite their large popularity in the HCI community [33].

In this context, represented by a growing scientific interest in smart rings for diverse application domains and contexts of use [30], but overall few replications of gesture elicitation studies [14], we contribute a replication of Gheran *et al.* [11]. Our results show that only 50% of the gestures discovered during the replication match those proposed by the participants of the original study, although we used a similar sample and the same analysis method; see Figure 1. Based on our results, we stress the importance of replications to confirm and consolidate—or contradict and refute—previous findings about user-defined gestures, and we draw implications for the theory and practice of gesture elicitation studies.

2 RELATED WORK

Relevant related work is represented by the scientific literature on gesture elicitation studies, for which we refer readers to Wobbrock *et al.*’s [37,38] seminal papers, Villarreal *et al.*’s [33] systematic literature review, and Vatavu and Wobbrock’s [32] critical overview and clarification of methodological aspects. According to the operational model for general end-user elicitation studies in HCI [32], the outcomes of such studies are represented by consensus sets of signs (e.g., consensus gesture sets for gesture elicitation studies) and/or characterizations of traits to understand user behavior with

interactive systems. However, the outcomes of any study need to be confirmed toward consolidating knowledge, but replications have been overall limited in HCI [16]. To foster replication of gesture elicitation studies, Gheran *et al.* [14] introduced RepliGES, which we illustrate in Figure 1 to position our work as a “type-5 replication.”

Despite the popularity of gesture elicitation studies [33], examination of user-defined ring gestures for input [11] or output [6] has been scarce. Gheran *et al.* [11] conducted the first end-user elicitation study of smart ring gestures motivated by the potential of ring-based input for a variety of application domains and contexts of use. We performed a forward snowballing procedure [39] starting from [11] and identified 87 references,¹ of which only two [2,13] were about gesture elicitation: Gheran *et al.* [13], a “type-9 empirical repurposing,” according to RepliGES terminology, where data from [11] were analyzed for a different goal, and Bardot *et al.* [2], who conducted a gesture elicitation study focused on a specific ring designed for input while grasping objects. In this context, design knowledge about smart ring gestures reflective of end users’ preferences remains very limited.

3 STUDY

We performed a replication of [11], the first gesture elicitation study conducted to examine users’ preferences for smart ring gestures. We chose a “type-5 replication,” according to RepliGES terminology (Figure 1), since it is the only replication type not yet conducted for gesture elicitation studies; see [14, p. 5:5]. Moreover, a type-5 replication enables verification of prior findings by being both exact [29] and strict [16], i.e., the same method is used as in the original study with gestures elicited from new participants.

3.1 Participants

We recruited 24 participants (15 male, 9 female) aged between 22 and 45 years old ($M=26.3$, $SD=5.7$). Both the size and gender structure of our sample were identical to those from [11, p. 625].

3.2 Design

Our study was a within-subjects design with two independent variables, replicating the design of Gheran *et al.* [11]:

- (1) **REFERENT**, nominal variable with five conditions representing various system functions common to a smart home environment: “turn TV on/off,” “start music player,” “turn volume up,” “turn lights on/off,” and “go to the next item in a list.” These referents are a subset of those used in [11]; see Subsection 3.3 for more details about our choice.
- (2) **NUM-RINGS**, ordinal variable with two conditions, *one-ring* and *two-ring* gestures, specifying two contexts of use corresponding to unimanual and bimanual gesture input, respectively, just as examined by Gheran *et al.* [11].

3.3 Procedure and Measures

We did our best to replicate all of the implementation details of the original study as closely as possible. Namely, we recruited a sample of participants of equal size and similar age and gender profiles, used

¹Using Google Scholar, query ran on October 15, 2022.

Table 1: Characteristics of the original gesture elicitation study and our replication.

	Original study [11]	Replication (this paper)
Participants	N=24 (15 male and 9 female) between 21 and 45 years old (M=27.5, SD=7.9 years)	N=24 (15 male and 9 female) between 22 and 45 years old (M=26.3, SD=5.7 years)
Apparatus	two Ring Zero devices, video recording	two Ring Zero devices, video recording
Design	within-subjects design with two independent variables: REFERENT and NUM-RINGS	within-subjects design with two independent variables: REFERENT and NUM-RINGS
REFERENT	fourteen referents, of which “TV on/off,” “start player,” “volume up,” “lights on/off,” “go to the next item in a list”	five referents: “TV on/off,” “start player,” “volume up,” “lights on/off,” “go to the next item in a list”
NUM-RINGS	<i>one-ring</i> and <i>two-ring</i> , randomized across participants	<i>one-ring</i> and <i>two-ring</i> , randomized across participants
Gesture set	672 elicited gestures, of which 240 for the five referents used in the replication study	240 elicited gestures corresponding to the five referents used for the replication
Measures	AR [31], THINKING-TIME (seconds), GESTURE-GOODNESS (10-point Likert scale rating)	AR [31], THINKING-TIME (seconds), GESTURE-GOODNESS (10-point Likert scale rating)

the same apparatus (two Ring Zero² devices), and presented the same instructions to our participants as in [11, p. 626]; see Table 1. Furthermore, an author of the original study³ analyzed the videos of the elicited gestures using the same procedure employed in the original study [11, pp. 626–627]. We chose only five referents for our replication, which received contrasting agreement rates in the original study, spanning a wide range, from the highest agreement rate (.225 for “next”) to the lowest (.040 for “lights on/off”); see [11, p. 627]. Furthermore, following [11, pp. 627–628], we classified the elicited gestures along four dimensions:⁴

- *Structure* characterizes the importance of hand movement and hand pose during gesture articulation with five categories: *buttons only*, *hand poses only*, *hand motion*, *hand poses & motion*, and *mixed locales*.
- *Complexity* differentiates between *simple* and *compound* gestures, where the latter can be decomposed into individually meaningful gestures.
- *Symmetry* describes how the two hands synchronize during the articulation of ring gestures, with four categories: *dominant unimanual*, *nondominant unimanual*, *symmetric bimanual*, and *asymmetric bimanual*.
- *Locale* specifies where gestures are performed: *on the ring*, *on other surface*, *in the air*, and *mixed locales*.

We also used the following measures from [11]:

- The Agreement Rate (AR) of a referent r is the proportion of pairs of participants in agreement over their gesture proposals for referent r out of the total number of pairs that could be in agreement [31]:

$$AR(r) = \frac{1}{N(N-1)} \sum_{i=1}^N \sum_{j=1, j \neq i}^N \delta_{i,j} \quad (1)$$

²<https://web.archive.org/web/20170309191236/http://ringzero.logbar.jp>

³The gesture elicitation study [11] had three authors, but only one performed the gesture clustering procedure to calculate agreement rates.

⁴Gheran *et al.* [11] also employed the *Nature* dimension, according to which ring gestures were classified as *symbolic*, *metaphorical*, and *abstract*. Since this dimension is less important for the implementation of a gesture recognizer, we did not pursue it in our replication; see more discussion in [22].

where N is the number of participants and $\delta_{i,j}$ evaluates to 1 if the gestures elicited from the i -th and j -th participants are equivalent or substantially similar, and to 0 otherwise. AR computes to a real value in the interval $[0, 1]$.

- THINKING-TIME represents the time, in seconds, needed by a participant to propose a gesture for referent r .
- GESTURE-GOODNESS represents participants’ subjective assessments, as a rating between 1 and 10, of how well the proposed gestures fit the referents (10 indicates the best fit). This measure was introduced by Wobbrock *et al.* [38] with a 7-point Likert scale but, for purposes of replication fidelity, we used the 10-point scale from [11].

We also employed several new measures to compare the results of our replication against those of the original study, as follows:

- DISTINCT-GESTURES represents the number of unique gestures, either from the original study or our replication.
- SHARED-PROPORTION-DISTINCT represents the proportion of distinct gestures found in both the original study and our replication. For example, if the original study identified 42 distinct gestures and the replication 49, of which 40 were also found in the original study, then SHARED-PROPORTION-DISTINCT is $40/49=81.6\%$.
- SHARED-PROPORTION-ALL represents the proportion of all gestures found in both the original study and the replication. Unlike for SHARED-PROPORTION-DISTINCT, repetitions are considered. For example, if the 40 distinct gestures shared between the two studies were proposed for 111 times (repetitions included) of the 120 gestures elicited during the replication, then SHARED-PROPORTION-ALL is $111/120=92.5\%$.

4 RESULTS

The average agreement rates reported for the five referents in the original study [11] were AR=.118 and .085 for the *one-ring* and *two-ring* conditions. In our replication, the average ARs were .090 and .065, representing low values of agreement [31,32], just like in the original study, and the relative differences in the agreement rates of the NUM-RINGS conditions were roughly the same, –23.7% and

Table 2: Results obtained in the original gesture elicitation study and our replication.

Criteria	Category	One-ring gestures			Two-ring gestures		
		Orig.	Replication	Replication vs. Original	Orig.	Replication	Replication vs. Original
Total number of elicited gestures		120	120	–	120	120	–
DISTINCT-GESTURES		47	57	$\chi^2_{(1)}=1.374, p=.241, n.s.$	63	67	$\chi^2_{(1)}=0.151, p=.698, n.s.$
SHARED-PROPORTION-DISTINCT		–	13/57=22.8%	–	–	15/67=22.3%	–
SHARED-PROPORTION-ALL		–	66/120=55.0%	–	–	60/120=50.0%	–
Average Agreement Rate (AR)		.118	.090	-23.7%	.085	.065	-23.5%
	AR (TV on/off)	.120	.069	-42.5%	.065	.036	-44.6%
	AR (start player)	.076	.025	-67.1%	.018	.025	+38.9%
	AR (volume up)	.130	.083	-36.2%	.062	.054	-12.9%
	AR (lights on/off)	.040	.029	-27.5%	.134	.054	-59.7%
	AR (next)	.225	.243	+8.0%	.145	.156	+7.6%
THINKING-TIME (s)		4.9	3.4	$t_{(46)}=1.678, p=.100, n.s.$	3.1	3.0	$t_{(46)}=0.121, p=.904, n.s.$
GESTURE-GOODNESS		7.9	8.0	$t_{(46)}=0, p=1, n.s.$	8.1	8.4	$t_{(46)}=1.288, p=.204, n.s.$
Structure [†]	<i>buttons only</i>	14	15	$p=1, n.s.$	12	13	$p=1, n.s.$
	<i>hand motion</i>	13	78	$p<.001$	48	82	$p<.001$
	<i>hand motion & poses</i>	67	18	$p<.001$	41	17	$p<.001$
	<i>hand poses</i>	8	3	$p=.217, n.s.$	5	4	$p=1, n.s.$
	<i>mixed locales</i>	18	6	$p=.018, n.s.$	14	4	$p=.027, n.s.$
Complexity [†]	<i>simple</i>	88	92	$p=.655, n.s.$	100	80	$p=.005$
	<i>compound</i>	32	28	$p=.655, n.s.$	20	40	$p=.005$
Symmetry [†]	<i>dominant unimanual</i>	117	94	$p<.001$	52	82	$p<.001$
	<i>nondominant unimanual</i>	0	2	$p=.478, n.s.$	7	0	$p=.021, n.s.$
	<i>symmetric bimanual</i>	3	13	$p=.020, n.s.$	44	19	$p<.001$
	<i>asymmetric bimanual</i>	0	11	$p=.002$	17	19	$p=.857, n.s.$
Locale [†]	<i>on the ring</i>	19	16	$p=.715, n.s.$	14	13	$p=1, n.s.$
	<i>in the air</i>	88	85	$p=.774, n.s.$	82	87	$p=.572, n.s.$
	<i>on other surface</i>	0	8	$p<.0125$	13	7	$p=.243, n.s.$
	<i>mixed locales</i>	13	11	$p=.830, n.s.$	11	13	$p=.830, n.s.$

[†] Bonferroni corrections were applied at the level of significance $\alpha=.05/n$, where n is the number of comparisons. For example, α was $.05/5=.01$ for the tests of proportions conducted for the *Structure* dimension, $.05/4=.0125$ for the *Locale* dimension, etc.

–23.5%, respectively (Table 2, row 5). To understand these results, we analyze the number of distinct gestures elicited in each study and the proportion of gestures shared between the two studies. But first, we examine the thinking times and gesture goodness ratings to understand similarities of behavior and perception of the participants from the two studies.

4.1 Thinking Time and Gesture Goodness

The average THINKING-TIME needed by the participants of the original study to propose gestures was 4.9s for the *one-ring* condition and 3.1s for *two-ring* [11, p. 629]. The average GESTURE-GOODNESS was 7.9 and 8.1 [11, p. 629], showing high confidence of the original participants in their gesture-to-referent mappings. The results of our replication were remarkably similar (Table 2, rows 11–12), and independent-samples t -tests did not reveal statistically significant differences between the mean THINKING-TIME (3.4s vs. 3.0s) and GESTURE-GOODNESS (8.0 vs. 8.4) measured in our replication compared to the original study. These results confirm the findings of [11], and highlight similarities between [11] and our replication in terms of participant behavior (THINKING-TIME) and perception of the gestures they proposed (GESTURE-GOODNESS).

4.2 Unique and Shared Gestures

The original study identified 47 distinct gestures of the 120 elicited in the *one-ring* condition and 63 distinct gestures of the 120 elicited in the *two-ring* condition. With new participants and the same gesture analysis procedure, we found 57 distinct gestures for *one-ring* and 67 for *two-ring*, respectively (Table 2, row 2). A Pearson chi-square test for the equality of proportions (Bonferroni corrections applied) did not find any statistically significant difference between the DISTINCT-GESTURES of the original study and our replication. The numbers of shared distinct gestures were only 13 and 15, respectively, resulting in a SHARED-PROPORTION-DISTINCT of approximately 22.0% for each of the NUM-RINGS conditions. When counting gesture repetitions, the values of SHARED-PROPORTION-ALL were 66/120=55% and 60/120=50%, respectively.

These results show that about half of the gestures elicited in our replication are new compared to the original study, and complement the dataset reported in [11]. Figure 2 illustrates a selection of these gestures. For example, one of our participants proposed a bimanual gesture, where the left hand swipes to the left and the right hand to the right, to “turn the volume up” (Figure 2a). The same participant proposed a *one-ring* gesture, where the finger touches an imaginary

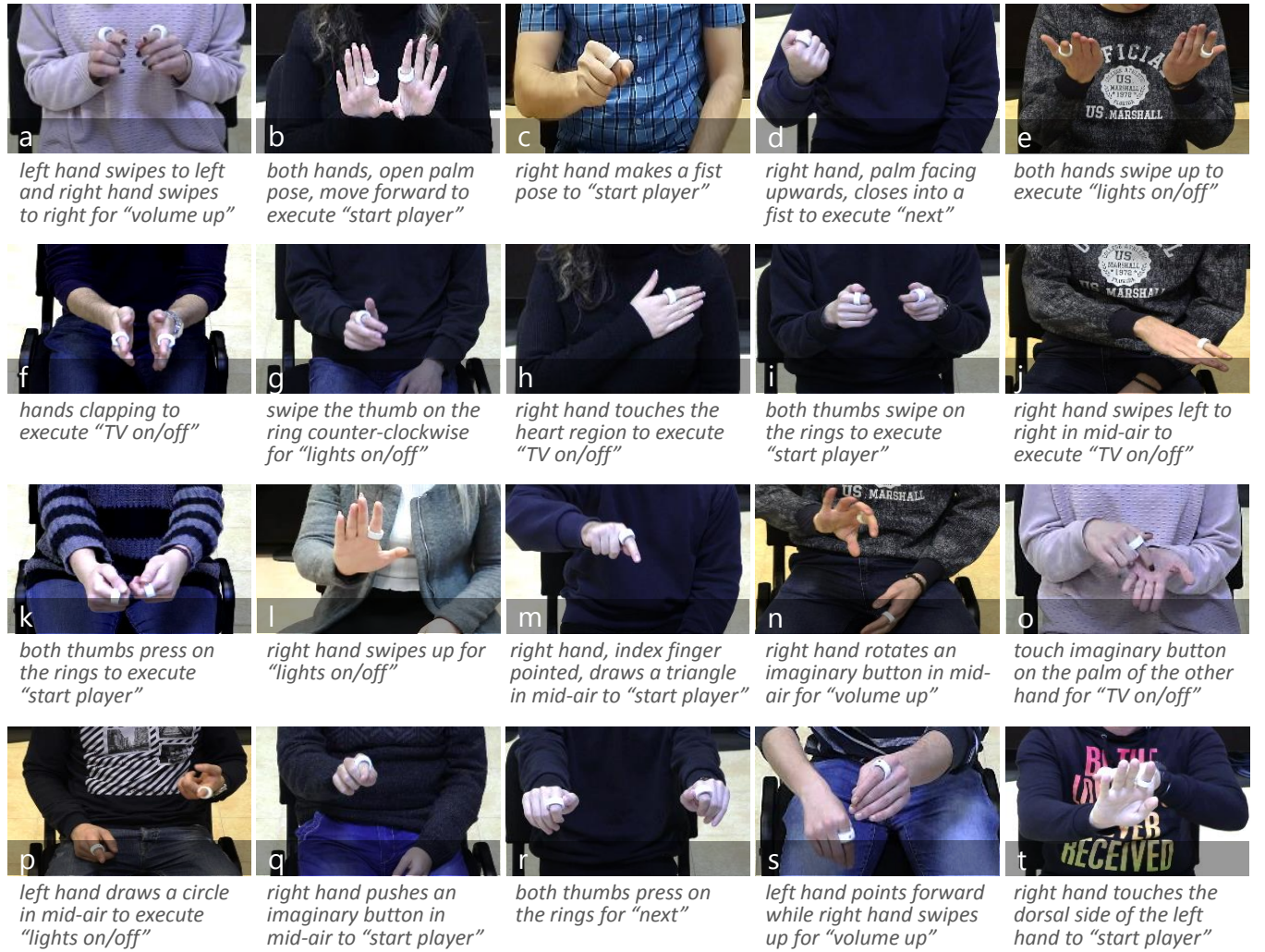


Figure 2: A selection of user-defined gestures from our replication, not discovered in the original study [11].

button on the opposite hand (Figure 2o), to “turn the TV on/off.” Another participant pushed both hands forward to “start the music player” (Figure 2b) and moved the hand wearing the smart ring close to the heart to “turn the TV on/off” (Figure 2h), which are two examples of body-referenced smart ring gestures not discovered in the original study. Furthermore, gestures similar to those proposed by the original participants were attributed to different referents by the participants from the replication, such as moving the finger up and down to “turn the lights on/off” (Figure 2l). It is not our goal to analyze these gestures in detail or to compare the consensus gesture sets between the two studies, especially since the agreement rates were small for both the original study and our replication. However, to enable further examination of such aspects, we provide the full list of elicited gestures as a companion dataset to our paper; see Subsection 5.2. The important takeaway is that our replication confirmed only 50% of the smart ring gestures found in the original end-user gesture elicitation study of Gheran *et al.* [11].

4.3 Gesture Properties

We used the *Structure*, *Complexity*, *Symmetry*, and *Locale* dimensions to analyze the gestures proposed by the new participants. Table 2 shows that most of the comparisons conducted between the original study and our replication resulted in non-significant differences (Bonferroni corrections applied) in terms of the proportion of the elicited gestures falling in the categories of these dimensions. For instance, the original study reported a number of 19 gestures performed *on the ring surface* in the *one-ring* condition, and our replication found 16 ($p=.715> .05/4=.0125$, *n.s.*). However, we also detected several statistically significant differences. These included the *hand motion* and *hand motion & poses* categories of the *Structure* dimension in both the *one-ring* and *two-ring* conditions ($p<.001$), and the *simple* and *compound* categories of the *Complexity* dimension in the *two-ring* condition ($p=.005$); see Table 2 for details. These results show that not only did our replication uncover new gestures compared to the original study, but those gestures

also have different characteristics in terms of *Structure*, *Complexity*, *Symmetry*, and *Locale* compared to the original ones.

5 DISCUSSION

We use our empirical findings to outline implications for replicating gesture elicitation studies, but also for the theory and practice of such studies. We also present limitations of our replication.

5.1 Implications for Gesture Elicitation Studies

The most important finding of our replication study is that we confirmed only half of the gestures found by Gheran *et al.* [11], while the other half represented new gestures with different articulation characteristics. One key implication of this finding is that conducting gesture elicitation studies with around 20-25 participants⁵ most likely limits the discovery of a representative set of user-defined gestures reflective of the preferences of the larger end-user population. On a more general note at the level of HCI research, a statement from one of the RepliCHI panels/workshops [36] corroborates on this aspect: “As a community, therefore, we do not value, facilitate, or reward replication in research, and often take the significant results of a single user study on 20 users to be true” (p. 463). Two implications follow: (i) replications of previously published gesture elicitation studies that employed around 20-25 participants (or less) need to be conducted to confirm (or contradict) their findings, and consolidate (or discover new) user-defined gestures, and (ii) more theoretical work is needed regarding the sample size of participants that are recruited for gesture elicitation studies.

The first implication strengthens Hornbæk *et al.*'s [16] recommendation, following the results of their literature review, to conduct more replications in the field of HCI in order to confirm, expand, or generalize the findings reported by earlier studies. In the specific context of gesture elicitation, the replication types of the RepliGES space [14], illustrated in Figure 1, can be used to inform new studies according to the goal, method, data, and population criteria. The RepliGES space was built on top of the replication literature [15,16,29,36], from which it adopted methods and provided terminology adaptations to speak directly to HCI practitioners and help identify replication types relevant for gesture elicitation studies. For example, repurposability (type-9 and type-10 in Figure 1) and reproducibility (type-2) can be implemented toward more discoveries about users' preferences for gesture input in contexts of use [8] other than those addressed in the original studies. Type-5 replications, as conducted in this work, are possible only when the same method of analysis can be applied in the same way, which means that the method must be described in sufficient detail in the original study to enable successful replication. Unfortunately, many gesture elicitation studies do not specify all of the implementation details, such as the context of use of their experimental setup, which makes replications difficult. Nevertheless, such replications are recommended, since space and place are known to influence how participants engage in a gesture elicitation study [1], and location and audience influence the social acceptability of gestures performed in public places [25]. Moreover, different user categories

may have different gesture preferences according to their age [4], dexterity and motor abilities [3], and cultural background [10,19]. In this context, instead of focusing exclusively on seeking originality, i.e., publish gesture elicitation studies for the newest application domains or interactive systems, incremental research in the form of replicating previous findings—e.g., agreement rates, consensus gesture sets, gesture articulation characteristics, etc.—is needed to consolidate our current knowledge of user-defined gestures.

The second implication recommends further work on the theoretical foundations of end-user elicitation studies, for which the impact of the sample of participants on the results should be carefully considered. A desirable outcome is a mathematical model to estimate the sample size needed to confidently discover a gesture set representative for the larger end-user population or, respectively, to draw meaningful conclusions about the traits of gesture articulation for understanding end-user behavior with interactive systems—the two outcomes of an end-user elicitation study [32]. The machine learning-inspired cross-validation approach of Tsandilas and Dragicevic [28], which estimates the guessability [37] error of sign-to-referent mappings for increasingly large sample sizes, can be used by HCI practitioners to decide when to stop collecting more gestures from more participants and, thus, is a good starting point for informing the development of such a model. Nielsen and Landauer's [23] approach, based on a mathematical model for the discovery of usability problems, may also prove helpful in this regard, whereas Villarreal *et al.*'s [33] systematic literature review of gesture elicitation studies represents a rich source of information about the characteristics of the samples used in previous studies.

5.2 Limitations

There are two limitations to our work. First, we elicited gestures for only a subset of the referents of the original study. Although this subset was enough to discover new gestures, not proposed in the original study, running a full replication with all of the referents is recommended toward more findings about replicating results about user-defined ring gestures, but also for identifying a smart ring gesture set that better reflects end users' preferences. Our choice of using fewer referents led to a shorter study compared to Gheran *et al.*'s [11], which allowed our participants to better recall their previously proposed gestures, whereas the original participants likely experienced higher physical and mental workload, e.g., more arm fatigue because of producing more gestures that were also more difficult to remember. For further examinations in this regard, we provide the full list of the elicited gestures as companion data to our paper at <http://www.eed.usv.ro/~vatavu>. Second, we proposed implications for gesture elicitation studies informed by our specific findings on ring gestures. Given that smart rings still represent new technology with which users are little or not familiarized, the low agreement rates observed in the original study and our replication may also be the direct result of technology novelty. Replications conducted for other application domains or interactive devices, e.g., multitouch surfaces, could confirm more of the original results.

6 CONCLUSION

We conducted a replication study about user-defined ring gestures in the context of a growing interest for smart rings and ring-based

⁵Villarreal *et al.* [33] reported, in their systematic literature review of 216 gesture elicitation studies, that the most frequent choice of sample size was 20 participants (36% of the published studies) and the average number of participants was 25 (p. 859).

input. We found that only half of the gestures of the original study were rediscovered. Our findings have implications for the replication of gesture elicitation studies in order to confirm and consolidate—or contradict and refute—their findings, but also for the theoretical foundations and best practices of running such studies.

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